Day5:
Computational Workflows for Molecular Dynamics Simulations of Polymers Using GROMACS

Course Duration: 5 Days (Online Live Sessions) — 5:00 to 6:30 PM IST
Instructor: **Dr. Harish Gudla** (Compular, UU)
Organized by: **MetaChem Academy** LLP

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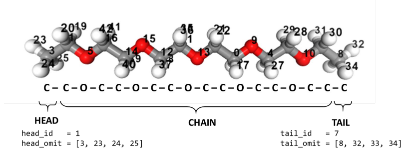
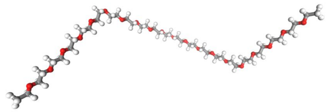
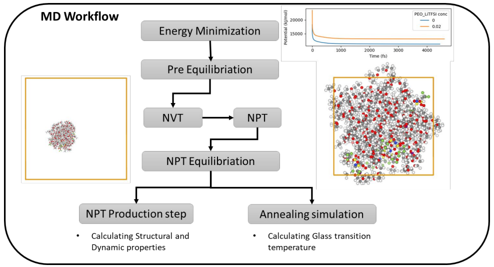
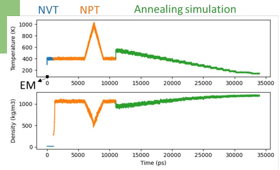
Workshop Schedule

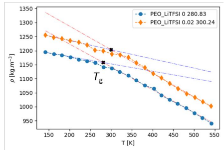


Date	Theory Session (5:00–5:30 PM IST)	Hands-On Session (5:30–6:30 PM IST)
10 Dec	Introduction to classical MD, force fields, and motivation for polymer simulations	Installing and setting up tools and Parameterizing monomers
11 Dec	Polymer chain architecture: HEAD/CHAIN/TAIL atom definitions and polymerization logic	Getting to jupyter notebooks and Parameterizing monomers Building polymer chains using parameterized monomers and preparing input files
12 Dec	Polymer chain architecture: HEAD/CHAIN/TAIL atom definitions and polymerization logic	Building polymer chains using parameterized monomers and preparing input files
15 Dec	GROMACS input formats and MD simulation workflows	Run first GROMACS MD on single polymer chain and perform initial analysis
16 Dec	Full MD simulation workflow from System Preparation to Production and Analysis of conformational and thermal properties	Production simulation and Performing annealing runs and calculating polymer properties: density, RMSD, Rg, and Tg









Day 5 Outline

- Recap
  - System Preparation in GROMACS
  - Equilibration steps

- Theory/Hands-on session (5:00-5:30 pm IST)
  - Production Simulation
  - Main Analysis
  - Polymer simulation applications

Classical Force Field

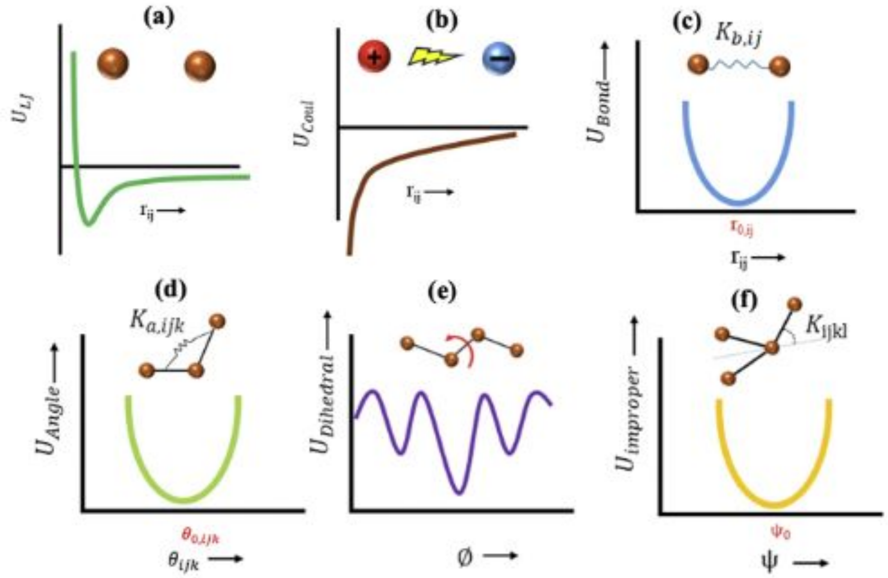
The functional form of a typical forcefield

$$U_{\text{total}} = U_{\text{LJ}} + U_{\text{Coul}} + U_{\text{Bond}} + U_{\text{Angle}} + U_{\text{Dihedral}} + U_{\text{Improper}}$$

Non-bonded interactions

$$U_{\text{LJ}} = 4\epsilon_{ij}\left[\left(\frac{\sigma_{ij}}{r_{ij}}\right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}}\right)^{6}\right],$$

$$U_{\text{Coul}} = \frac{q_i q_j}{4\pi\epsilon_1\epsilon_0 r_{ij}},$$



Bonded interactions

$$U_{\text{Bond}} = K_{b,ij}(r_{ij} - r_{0,ij})^2,$$

$$U_{\text{Angle}} = K_{a,ijk}(\theta_{ijk} - \theta_{0,ijk})^2,$$

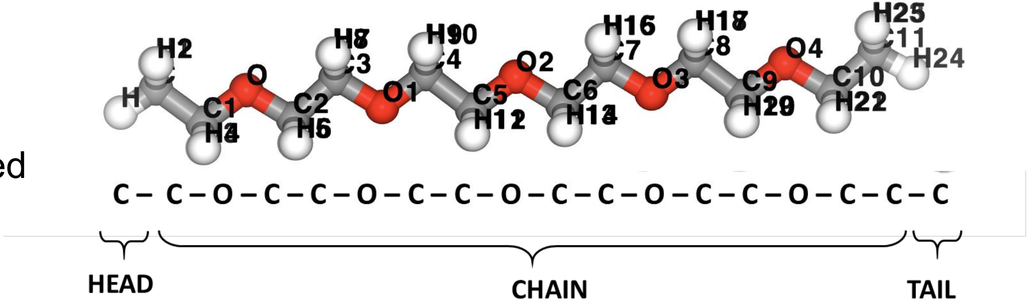
$$U_{\text{Dihedral}} = \frac{1}{2}K_1[1+\cos(\phi)] + \frac{1}{2}K_2[1-\cos(2\phi)] + \frac{1}{2}K_3[1+\cos(3\phi)] + \frac{1}{2}K_4[1-\cos(4\phi)],$$

$$U_{\text{improper}} = K_{ijkl}(\psi - \psi_0)^2.$$

Chem. Commun., 2024, 60, 6093–6129

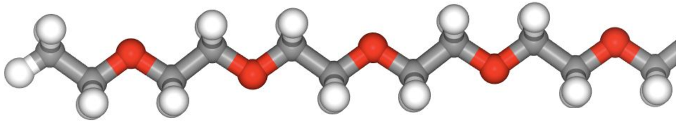
Polymer chain architecture PEO

head_id is the Atom label where the Head end group is connected

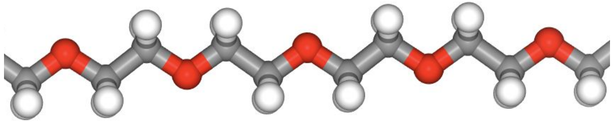



```
head_id   = 1
head_omit = [3, 23, 24, 25]
```

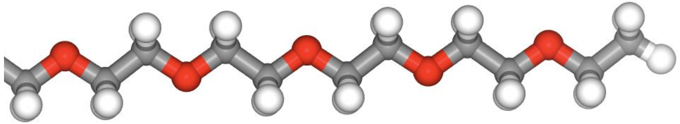
```
tail_id   = 7
tail_omit = [8, 32, 33, 34]
```



Head (HTP)



Chain (PEO)

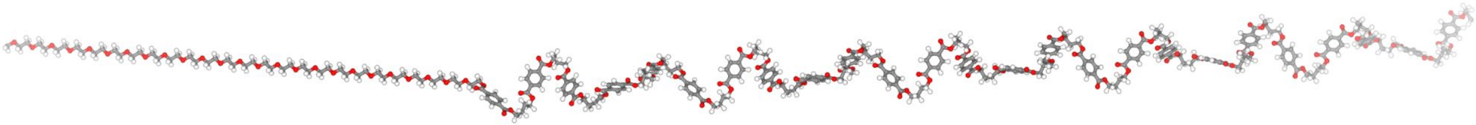


Tail (TPT)

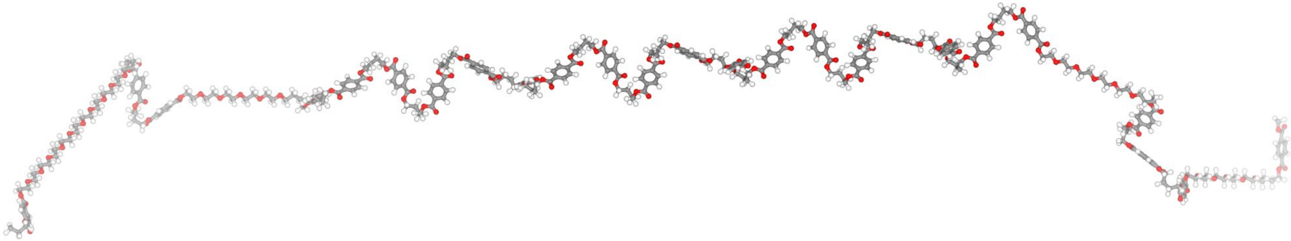
Polymer chain building with tleap

```
generate_polymer_tleap(
        mol_names=["PEO", "PET"],    # List of monomers to be present in polymer chain
        n_mono_repeat=[5, 1],        # List of monomers present in parameterised repeat unit
        n_mono_pol=[25, 25],         # Desired number monomers present in polymer chain
        copolymer_type="random",     # Type of polymer chain homo, block, random
        head_group="PEO",
        tail_group="PET"
```

Block co-polymer PEO-PET

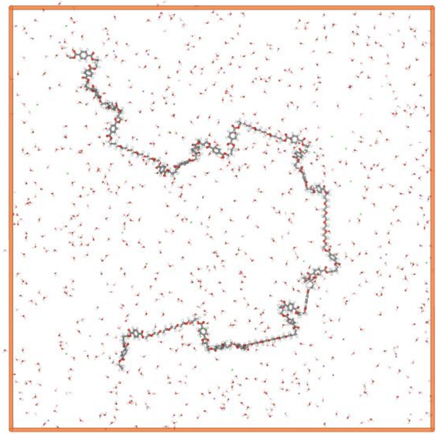


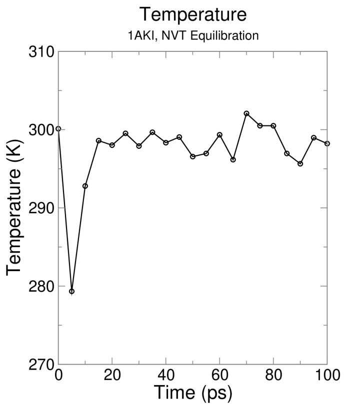
Random co-polymer PEO-PET

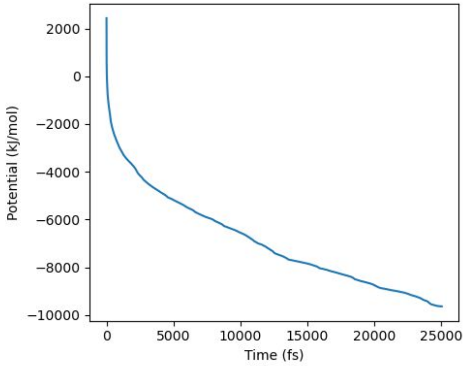


Main Stages of a MD simulation

- System Preparation
- Minimization/Relaxation
- Equilibration
- Production
- Analysis





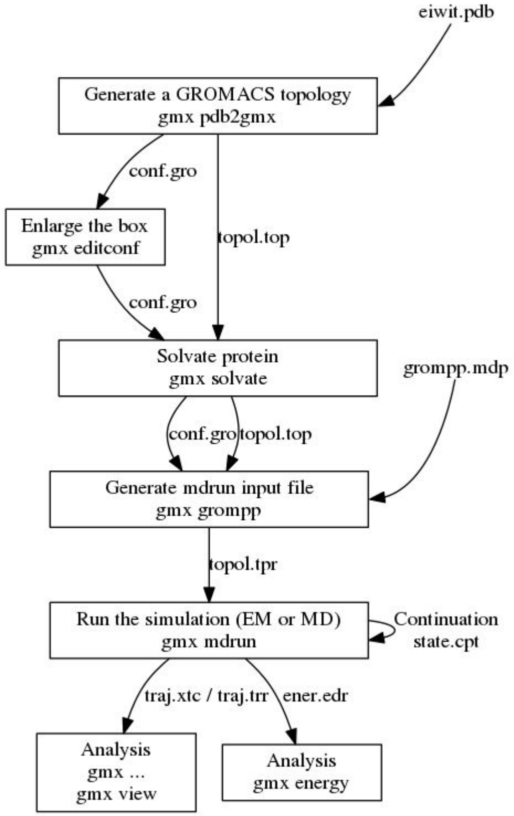


GROMACS command-line

```
$gmx command -options files/values/strings
```

- pdb2gmx
- editconf
- genbox
- solvate
- grompp
- mdrun
- energy
- ....

```
OPTIONS

Options to specify input files:

 -f      [<.gro/.g96/...>]  (eiwit.pdb)
           Structure file: gro g96 pdb brk ent esp tpr

Options to specify output files:

 -o      [<.gro/.g96/...>]  (conf.gro)
           Structure file: gro g96 pdb brk ent esp
 -p      [<.top>]           (topol.top)
           Topology file
 -i      [<.itp>]           (posre.itp)
           Include file for topology
 -n      [<.ndx>]           (clean.ndx)      (Opt.)
           Index file
 -q      [<.gro/.g96/...>]  (clean.pdb)      (Opt.)
           Structure file: gro g96 pdb brk ent esp

Other options:

 -chainsep <enum>           (id_or_ter)
           Condition in PDB files when a new chain should be started (adding
           termini): id_or_ter, id_and_ter, ter, id, interactive
 -merge  <enum>             (no)
           Merge multiple chains into a single [moleculetype]: no, all,
           interactive
 -ff     <string>           (select)
           Force field, interactive by default. Use -h for information.
 -water  <enum>             (select)
           Water model to use: select, none, spc, spce, tip3p, tip4p, tip5p,
           tips3p
```




```
$gmx pdb2gmx -h
```

To get help regarding the command's usage, description and options.

GROMACS formats

Input/Output File Formats:

- **.gro**: GROMACS coordinate file containing atomic positions and box dimensions.
- **.top**: Topology file describing the molecular structure, force field parameters, and system composition.
- **.mdp**: Molecular dynamics parameter file specifying simulation settings.

Output File Formats

- **.log**: Log file with simulation details and Energy contributions.
- **.tpr**: Portable binary run input file generated by gmx grompp (contains all info for a simulation run).
- **.edr**: Energy file containing simulation energy data.
- **.trr/.xtc**: Trajectory files (full-precision and compressed, respectively).

Main GROMACS Commands Used:

gmx editconf: Defines the simulation box and manipulates coordinates.
gmx solvate: Adds solvent (e.g., water) to the simulation box.
gmx genion: Adds ions to the system for neutralization and ionic strength.
gmx energy: Extracts energy and thermodynamic properties from simulation output.
gmx grompp: Preprocesses input files and creates the .tpr run file.
gmx mdrun: Runs the actual MD simulation or energy minimization.

System Preparation

Minimization

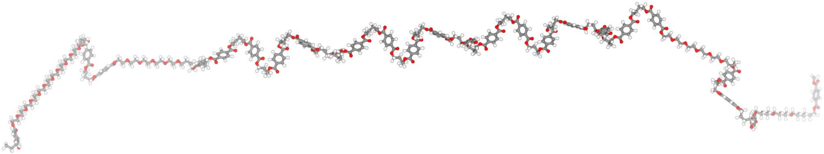
Equilibration

Production

Analysis

Initial Configuration files

polymer.gro
(Coordinates file)



```
GROningen MAchine for Chemical Simulation
  875
    1PET     C9    1   0.354   0.142   0.000
    1PET     H6    2   0.397   0.184   0.091
    1PET     H7    3   0.387   0.197  -0.088
```

topol.top
(Force field parameters) - atomtypes, atom/bond/pairs/dihedral/improper details

```
[ defaults ]
; nbfunc        comb-rule       gen-pairs       fudgeLJ  fudgeQQ
1               2               yes             0.5          0.83333333

[ atomtypes ]
; name     at.num     mass     charge ptype  sigma        epsilon
c3              6  12.010000  0.00000000  A     0.33977095       0.4510352
h1              1   1.008000  0.00000000  A     0.24219973       0.0870272
os              8  16.000000  0.00000000  A     0.31560978       0.3037584
c               6  12.010000  0.00000000  A     0.33152123       0.4133792
o               8  16.000000  0.00000000  A     0.30481209       0.6121192
ca              6  12.010000  0.00000000  A     0.33152123       0.4133792
ha              1   1.008000  0.00000000  A     0.26254785       0.0673624
hc              1   1.008000  0.00000000  A      0.2600177       0.0870272

[ moleculetype ]
; Name            nrexcl
system1          3

[ atoms ]
;   nr       type  resnr residue  atom   cgnr     charge       mass  typeB    chargeB      massB
; residue    1 PET rtp PET q 0.0
     1         c3      1    PET     C9      1 0.12984200  12.010000   ; qtot 0.129842
     2         h1      1    PET     H6      2 0.05392300   1.008000   ; qtot 0.183765
     3         h1      1    PET     H7      3 0.05392300   1.008000   ; qtot 0.237688
     4         os      1    PET     O2      4 -0.43055400  16.000000   ; qtot -0.192866
     5          c      1    PET     C8      5 0.65144800  12.010000   ; qtot 0.458582
```

```
[ system ]
; Name
Generic title in water

[ molecules ]
; Compound        #mols
system1              1
```

Initial pre-equilibration of single polymer

System Preparation

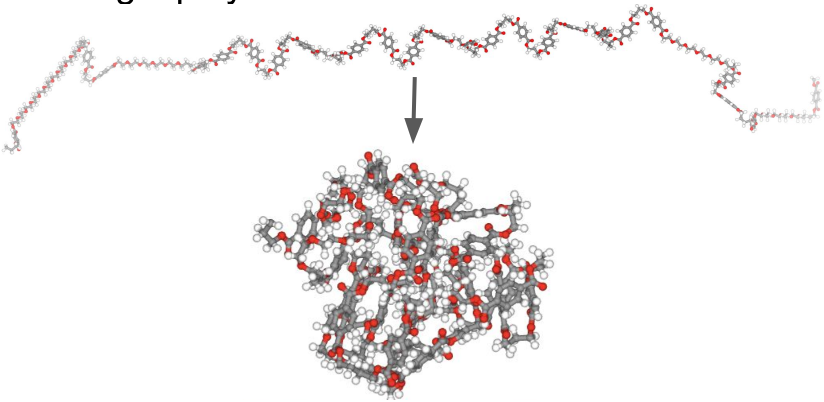
Minimization

Equilibration

Production

Analysis

polymer_relax.gro
(Coordinates file)

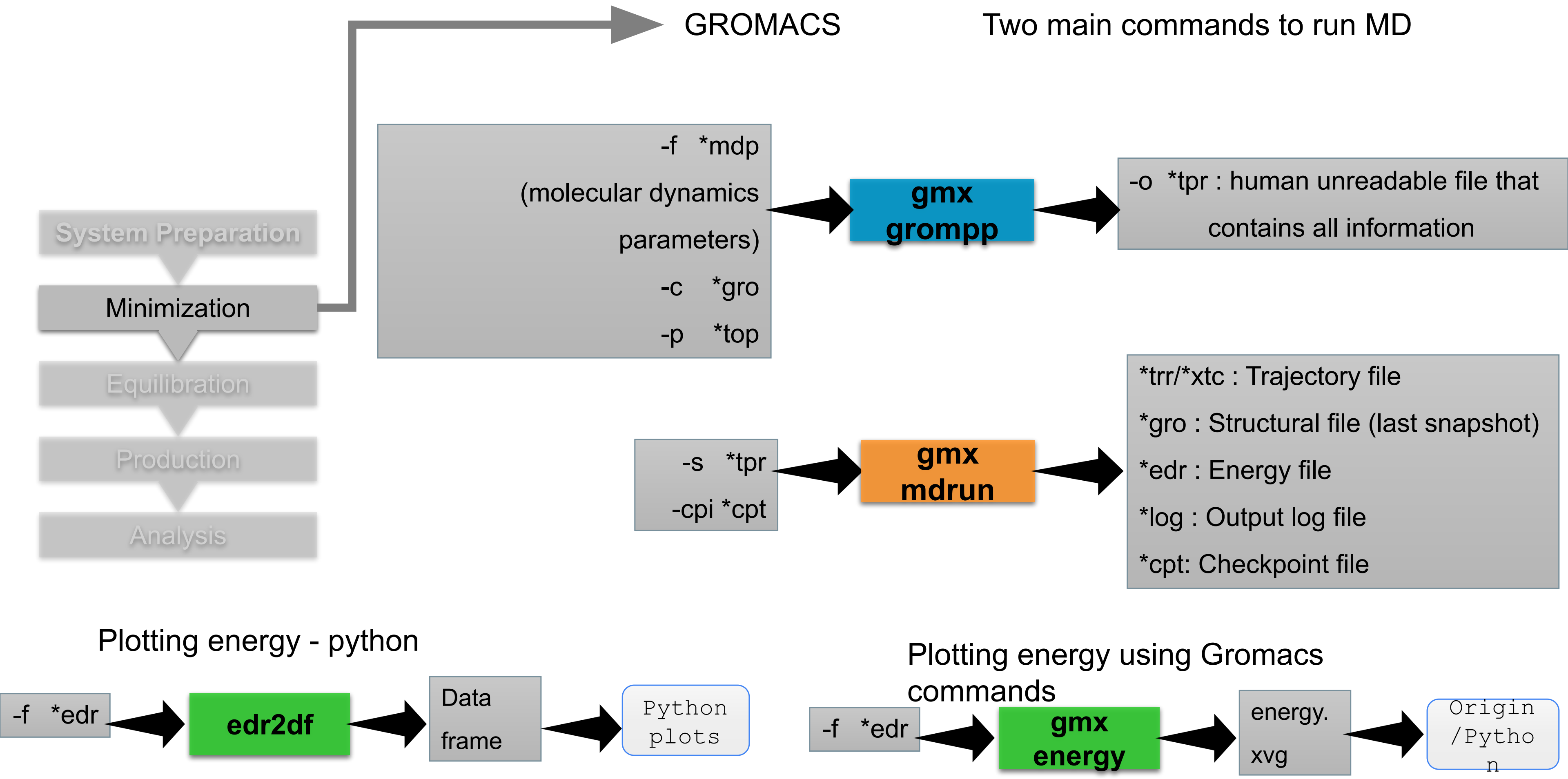


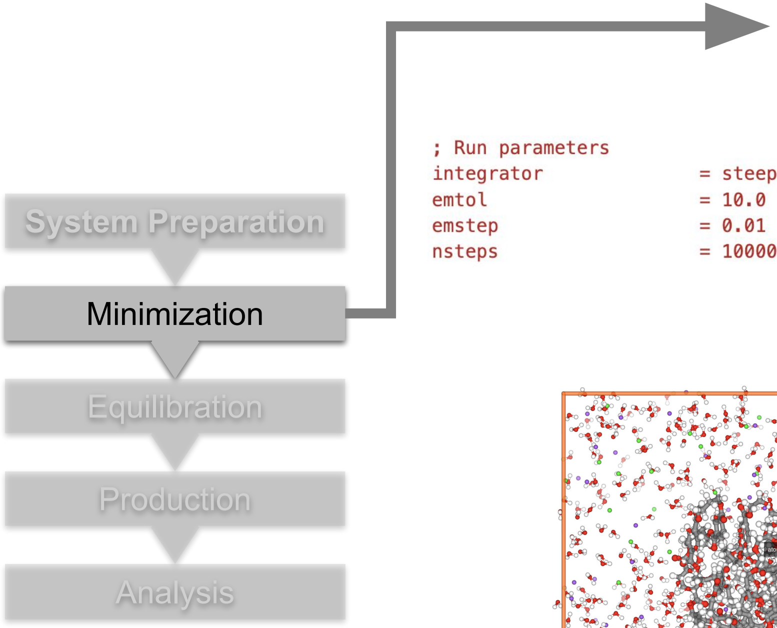
topol.top
(Force field parameters) - atomtypes, atom/bond/pairs/dihedral/improper details

```
[ system ]
; Name
Generic title in water

[ molecules ]
; Compound        #mols
system1               1
```

→ **gmx solvate** → **gmx genion** →

```
[ system ]
; Name
Generic title in water

[ molecules ]
; Compound        #mols
system1               1
SOL          800
NA                  100
CL                  100
..
```

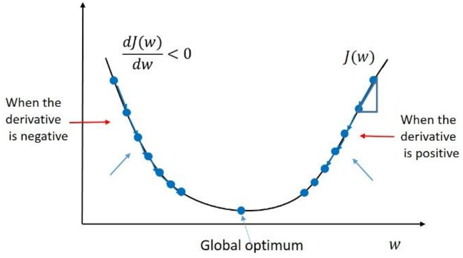

GROMACS
Two main commands to run MD
System Preparation
Minimization
Equilibration
Production
Analysis
-f *mdp
(molecular dynamics
parameters)
-c *gro
-p *top
gmx
grompp
-o *tpr : human unreadable file that
contains all information
-s *tpr
-cpi *cpt
gmx
mdrun
*trr/*xtc : Trajectory file
*gro : Structural file (last snapshot)
*edr : Energy file
*log : Output log file
*cpt: Checkpoint file
Plotting energy - python
-f *edr
edr2df
Data
frame
Python
plots
Plotting energy using Gromacs
commands
-f *edr
gmx
energy
energy.
xvg
Origin
/Pytho
n

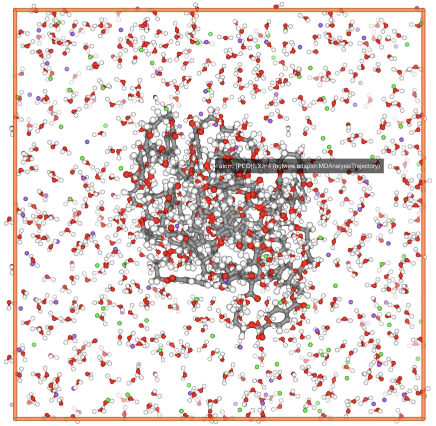


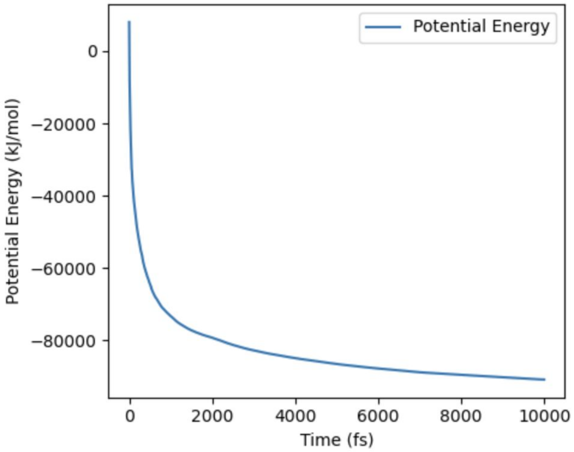
Energy minimization step

-using steepest descent algorithm

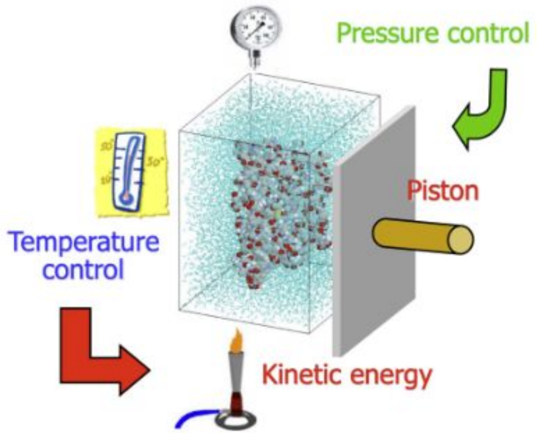
```
; Run parameters
integrator              = steep         ; Algorithm for energy minimization
emtol                   = 10.0          ; Convergence criterion (kJ/mol/nm)
emstep                  = 0.01          ; Step size for minimization
nsteps                  = 10000         ; Maximum number of steps
```





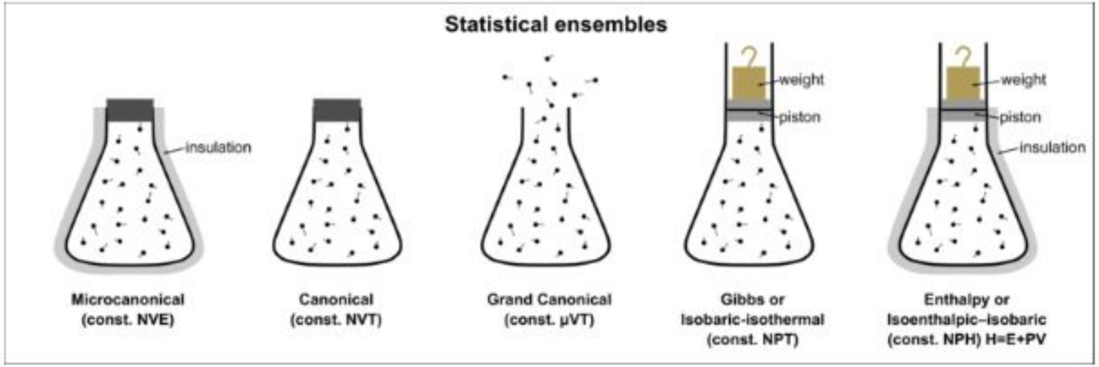


What are Statistical ensembles?



Statistical ensembles describe how **macroscopic conditions are controlled in molecular simulations** while particles evolve according to Newton's laws.

- NVE (Microcanonical ensemble)
  - Number of particles (N), volume (V), and total energy (E) are conserved.
- NVT (Canonical ensemble)
  - Number of particles (N), volume (V), and temperature (T) are fixed.
- NPT (Isothermal–Isobaric ensemble)
  - Number of particles (N), pressure (P), and temperature (T) are fixed.



NVT vs NPT ensemble

Thermostats

Velocity rescaling to achieve the target temperatures

$$T = \frac{< \sum_{i=1}^{N} m_i v_i^2 >}{3Nk_B T}$$

Simple velocity rescaling:

$$\lambda = \sqrt{\frac{T}{T_{target}}}$$

```
; Turning on thermostat
Tcoupl                 =   berendsen
tau_t                  =   0.01
tc_grps                =   System
ref_t                  =   298
```

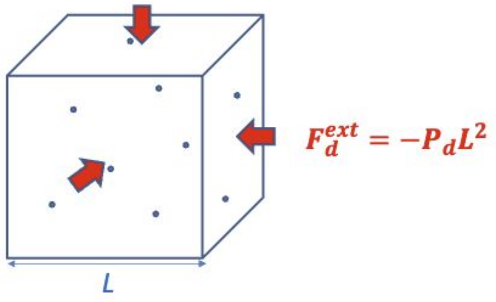
Common coupling algorithms:

- Berendsen (weak-coupling, equilibration)
- Nosè-Hoover thermostat/barostats
- Bussi-Donadio-Parrinello (v-rescale, production)

Barostats

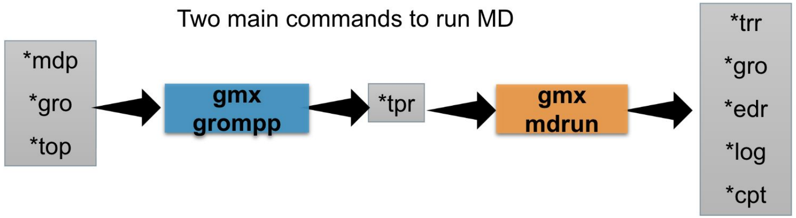
mostly implemented together with a thermostat so that both P and T are conserved along the course of our simulation

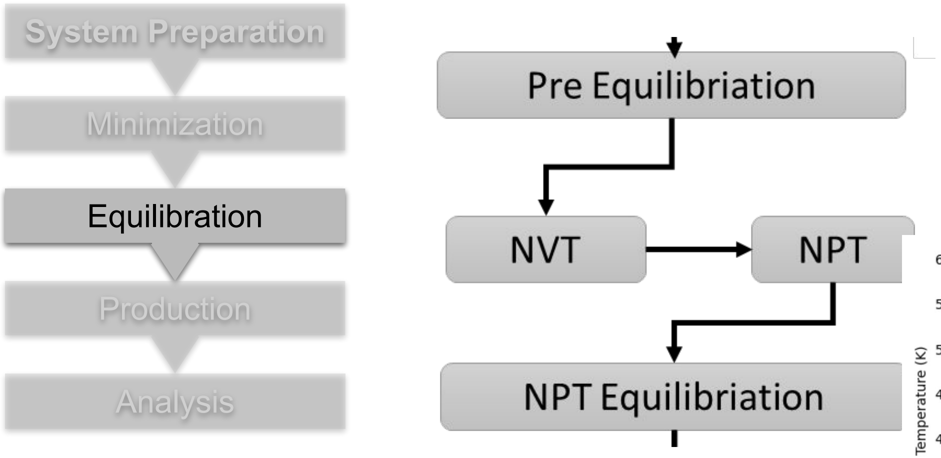
$$\frac{\partial P(t)}{\partial t} = \frac{1}{\tau}(P_{bath} - P(t))$$

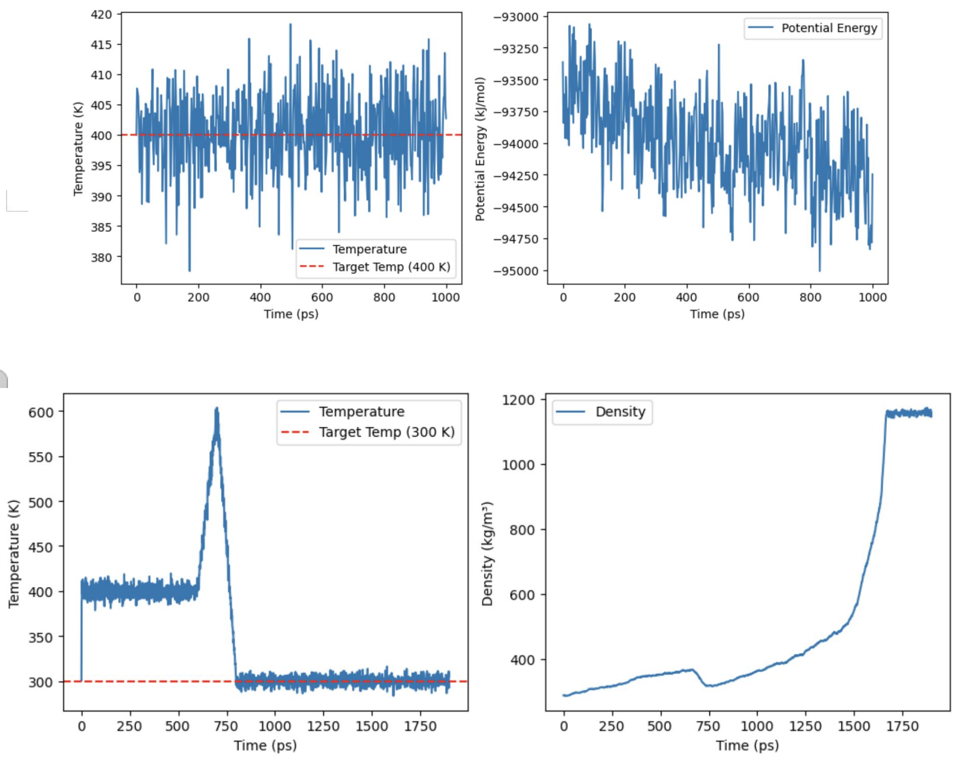



```
; Turn on barostat
Pcoupl                 =   berendsen
pcoupltype             =   isotropic
tau_p                  =   2.0
compressibility        =   4.4e-5
ref_p                  =   1.0
```

Two main commands to run MD







NVT <-> NPT Equilibration steps are repeated till density of the simulation is converged

System Preparation

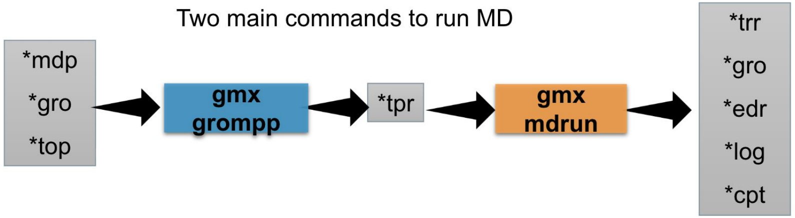
Minimization

Equilibration

Production

Analysis

Two main commands to run MD



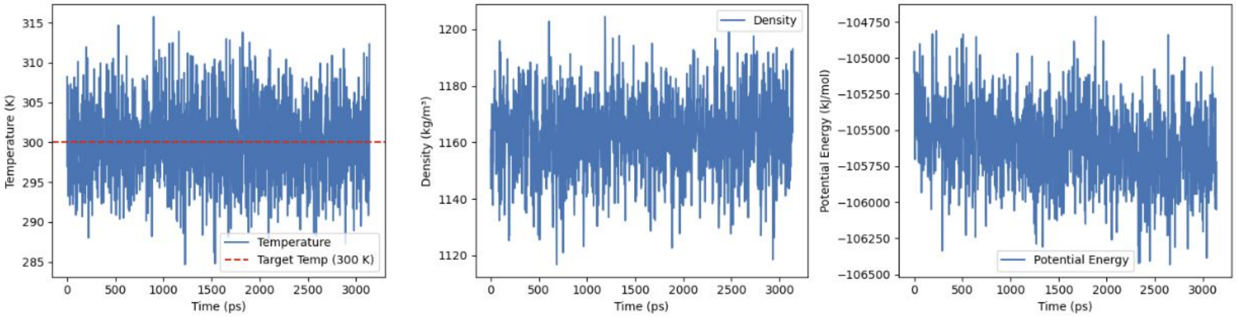
NPT Production step

- Calculating Structural and Dynamic properties

Annealing simulation

- Calculating Glass transition temperature

For polymer systems, production runs require >100ns simulations



Day 5 Outline

- Recap
  - System Preparation in GROMACS
  - Equilibration steps

- Theory/Hands-on session (5:00-5:30 pm IST)
  - Production Simulation
  - Main Analysis
  - Polymer simulation applications

- **Root Mean Squared Deviation (RMSD)**:
  - Analyze the structural stability of the polymer chain.
  - Numerical difference between the two structures, either the simulated protein with its crystal structure or the evolution of polymer structure compared to initial configuration

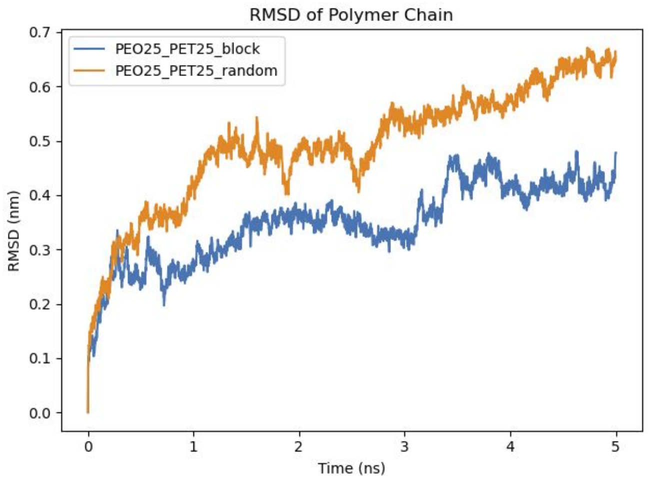
$$\mathrm{RMSD} = \sqrt{\frac{\sum_{i=1}^{N} (x_i - \hat{x}_i)^2}{N}}$$

RMSD = root-mean-square deviation
i = variable i
N = number of non-missing data points
x_i = actual observations time series
$\hat{x}_i$ = estimated time series

System Preparation
Minimization
Equilibration
Production
Analysis

*tpr *trr/*xtc → **gmx rms** → rmsd.xvg → Python plot

```
Command line:
  gmx rms -s ../prod-npt/prod-npt.tpr -f ../prod-npt/prod-npt.xtc -o rmsd.xvg -tu ns -b 1

Reading file ../prod-npt/prod-npt.tpr, VERSION 2025.2-conda_forge (single precision)
Reading file ../prod-npt/prod-npt.tpr, VERSION 2025.2-conda_forge (single precision)
Select group for least squares fit
Group     0 (         System) has  3483 elements
Group     1 (          Other) has   875 elements
Group     2 (            PEO) has   175 elements
Group     3 (            PET) has   700 elements
Group     4 (             NA) has    98 elements
Group     5 (             CL) has    98 elements
Group     6 (          Water) has  2412 elements
Group     7 (            SOL) has  2412 elements
Group     8 (      non-Water) has  1071 elements
Group     9 (            Ion) has   196 elements
Group    10 ( Water_and_ions) has  2608 elements
Select a group: 2
Selected 2: 'PEO'
Select group for RMSD calculation
Group     0 (         System) has  3483 elements
Group     1 (          Other) has   875 elements
Group     2 (            PEO) has   175 elements
Group     3 (            PET) has   700 elements
Group     4 (             NA) has    98 elements
Group     5 (             CL) has    98 elements
Group     6 (          Water) has  2412 elements
Group     7 (            SOL) has  2412 elements
Group     8 (      non-Water) has  1071 elements
Group     9 (            Ion) has   196 elements
Group    10 ( Water_and_ions) has  2608 elements
Select a group: 2
Selected 2: 'PEO'
Last frame       2000 time    5.000

Back Off! I just backed up rmsd.xvg to ./#rmsd.xvg.1#

GROMACS reminds you: "Bring Out the Gimp" (Pulp Fiction)
```



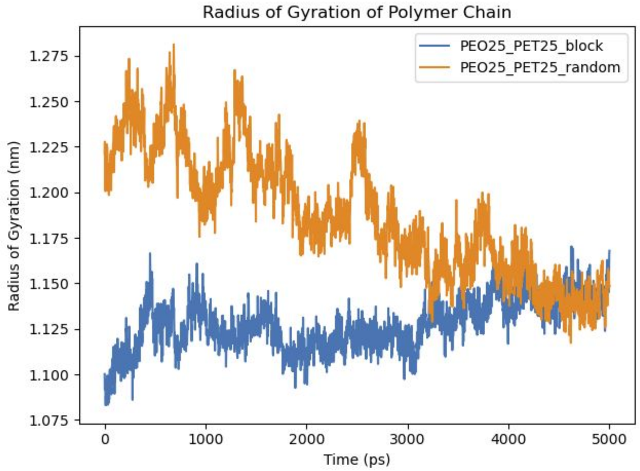
- **Radius of gyration (Rg)**:
  - Calculate the radius of gyration of the polymer chain to analyze its compactness over time.
  - Other similar properties, end-to-end distance, radius of rotation, hydrodynamic radius, and mass radius

$$R_{g,x} = \left(\frac{\sum_i \left(r_{i,y}^2 + r_{i,z}^2 \right) m_i}{\sum_i m_i} \right)^{\frac{1}{2}}$$

System Preparation → Minimization → Equilibration → Production → **Analysis**

*tpr, *trr/*xtc → **gmx gyrate** → gyrate.xvg → Python plot

```
Command line:
  gmx gyrate -s ../prod-npt/prod-npt.tpr -f ../prod-npt/prod-npt.xtc -o gyrate.xvg

Reading file ../prod-npt/prod-npt.tpr, VERSION 2025.2-conda_forge (single precision)
Reading file ../prod-npt/prod-npt.tpr, VERSION 2025.2-conda_forge (single precision)
Available static index groups:
 Group  0 "System" (3483 atoms)
 Group  1 "Other" (875 atoms)
 Group  2 "PEO" (175 atoms)
 Group  3 "PET" (700 atoms)
 Group  4 "NA" (98 atoms)
 Group  5 "CL" (98 atoms)
 Group  6 "Water" (2412 atoms)
 Group  7 "SOL" (2412 atoms)
 Group  8 "non-Water" (1071 atoms)
 Group  9 "Ion" (196 atoms)
 Group 10 "Water_and_ions" (2608 atoms)
Specify a selection for option 'sel'
(Select group to compute gyrate radius):
(one per line, <enter> for status/groups, 'help' for help)
> 1
Selection '1' parsed
Reading frame    2000 time 4000.000
Analyzed 2501 frames, last time 5000.000

GROMACS reminds you: "My Heart is Just a Muscle In a Cavity" (F. Black)
```



Polymer dynamics from MD simulations

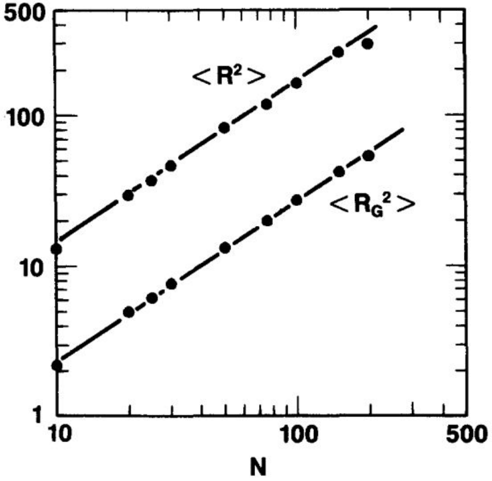
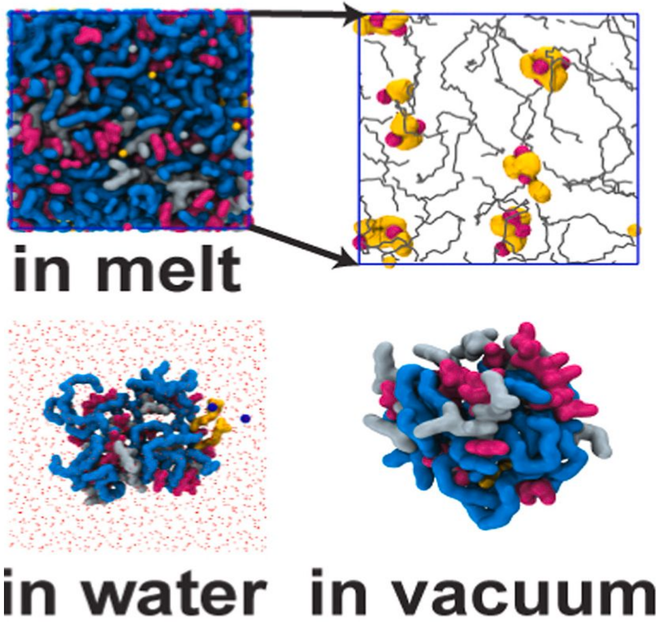
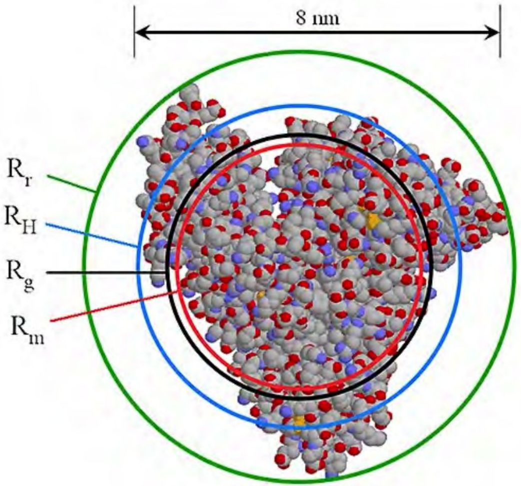


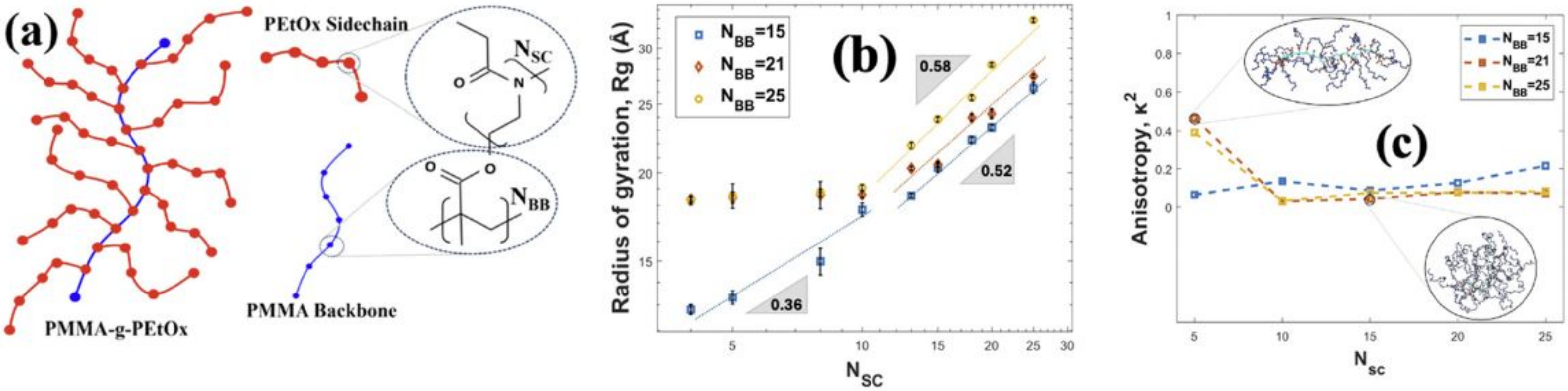
FIG. 1. Mean square end–to–end distance $\langle R^2(N) \rangle$ and radius of gyration $\langle R_G^2(N) \rangle$ vs N for $10 \leqslant N \leqslant 200$. The lines give the expected slope of 1.





Polymer 265 (2023) 125503

Bottlebrush polymers MD simulations



"the radius of gyration (Rg) of the BBPs scale with the length (or the number of monomers) of the side chains"

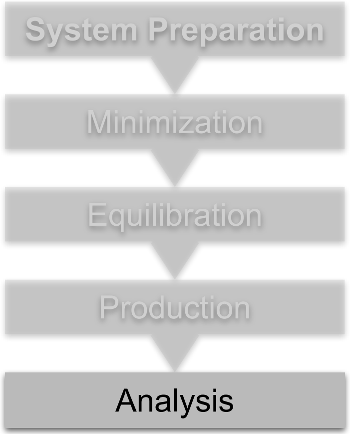
"as the side chain length is increased, the BBP transitioned from a rod-like shape to a sphere-like shape"

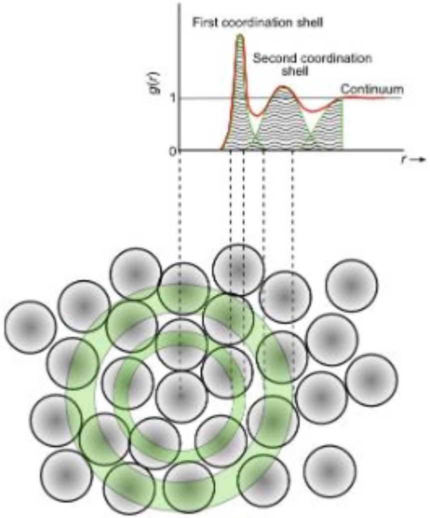
Langmuir, 2024, 40(6), 2946–2956

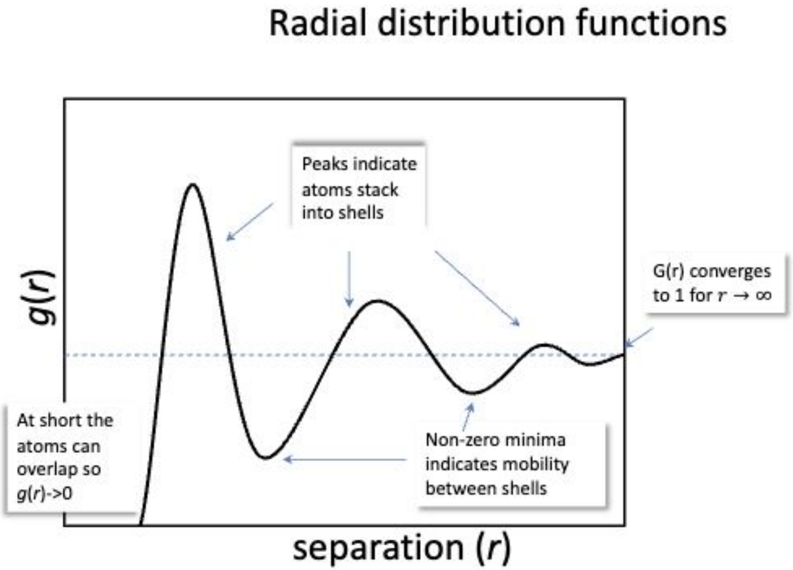
- **Radial distribution functions (RDF)**
  - Describes the probability of finding water/ions molecules at a certain distance from the polymer chains, providing insights into the local structure, hydrogen bonding, and confinement effects within the material.

computing g(r)

$$g(r) = \frac{< dn_r >}{4\pi r^2 dr * \rho}$$







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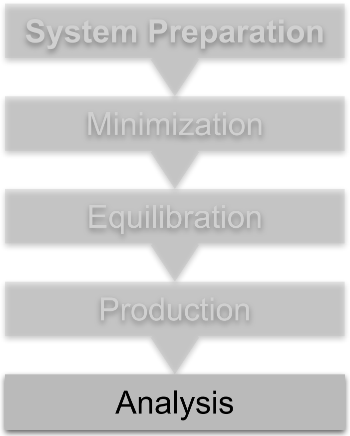
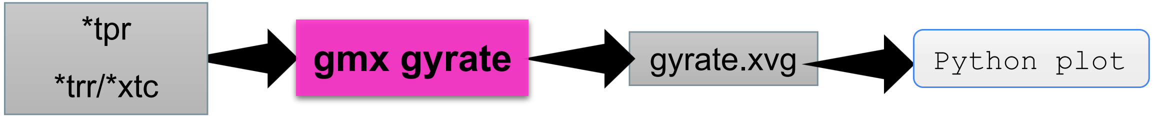
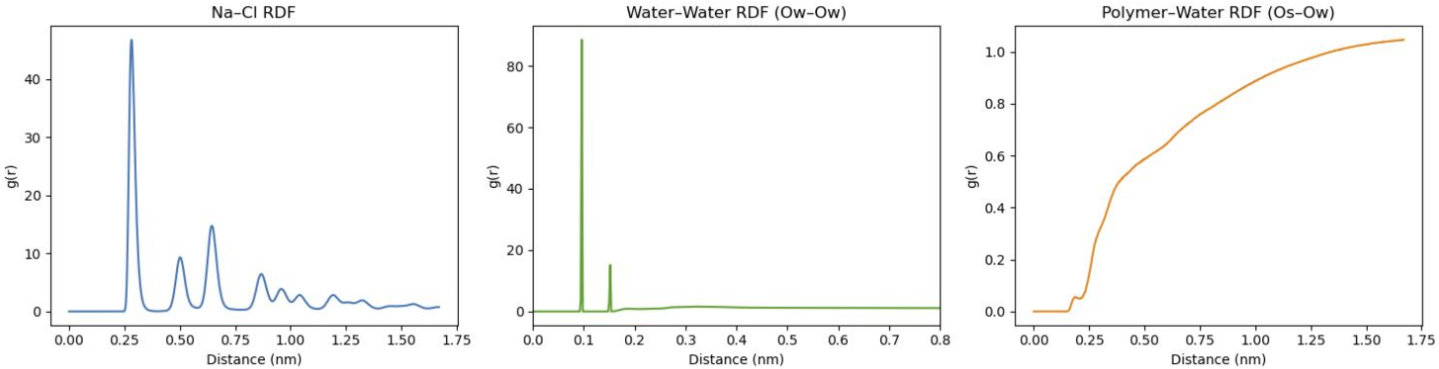
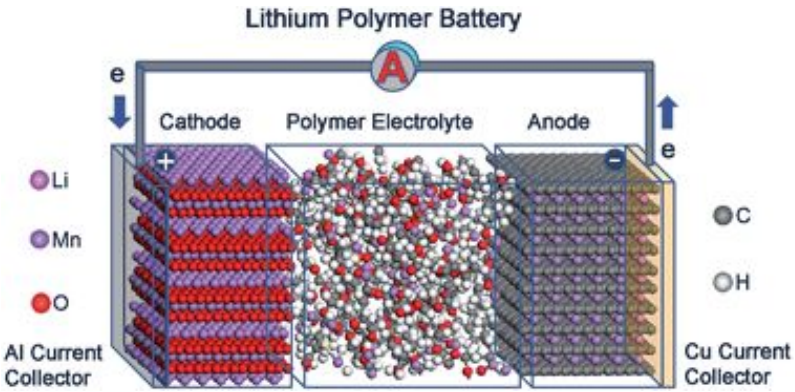






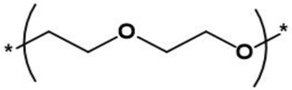
Polymer electrolytes in Li-ion batteries

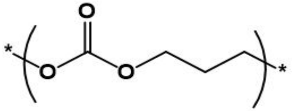


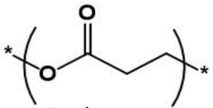
- ✔ Improved safety features
- ✔ Excellent flexibility and processability
- ✔ Suppressing dendrite growth

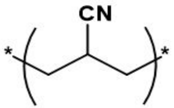
- ☐ Low ionic conductivity

Electrolyte	Ionic conductivity	Mechanical strength	Price	Safety
Liquid	1	4	3	4
Gel	2	3	2	3
Polymer	4	2	1	2
Ceramic	3	1	4	1

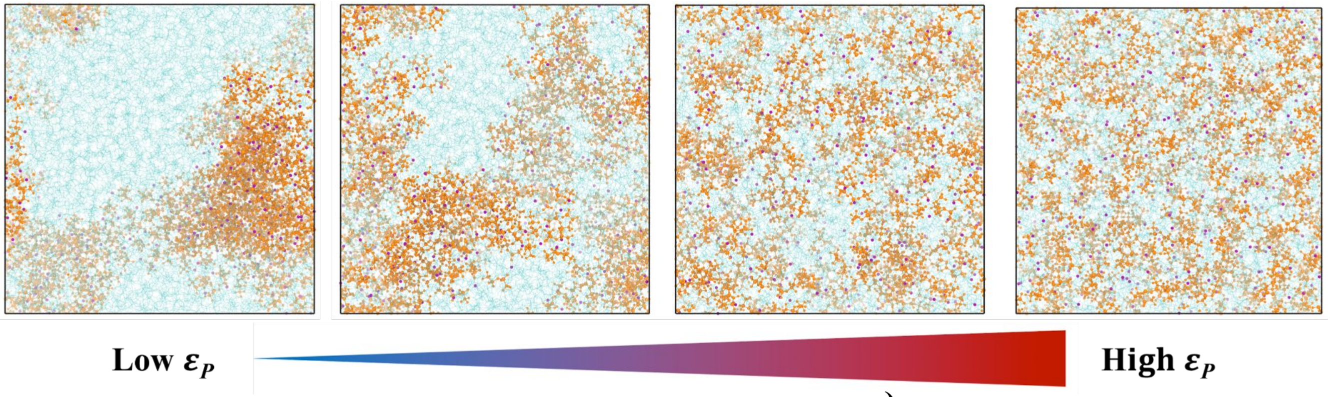

Poly(ethylene oxide) (PEO)

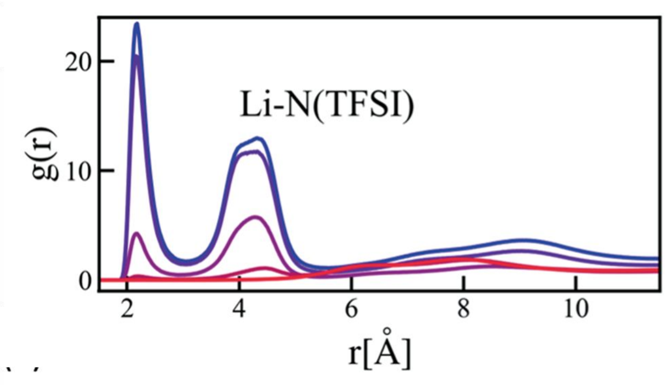

Polycarbonates

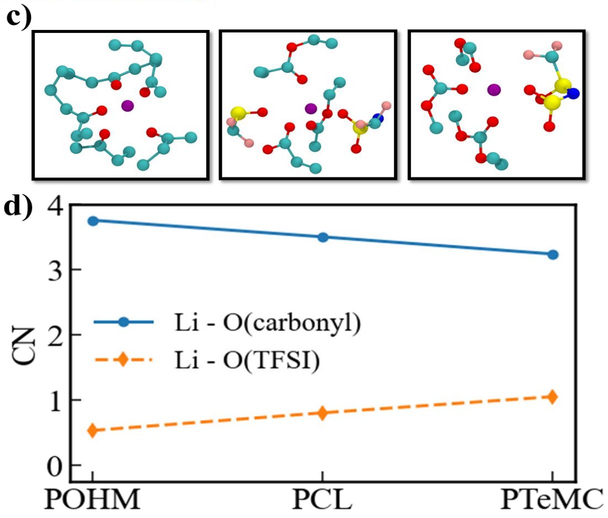

Polyesters


Polyacrylonitrile

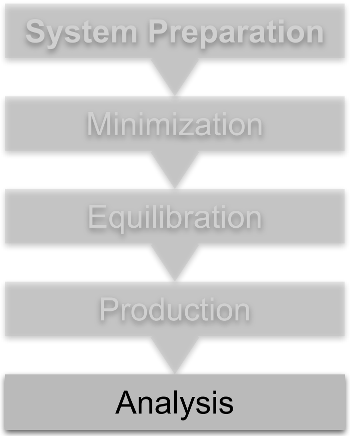
Polymer electrolytes in Li-ion batteries

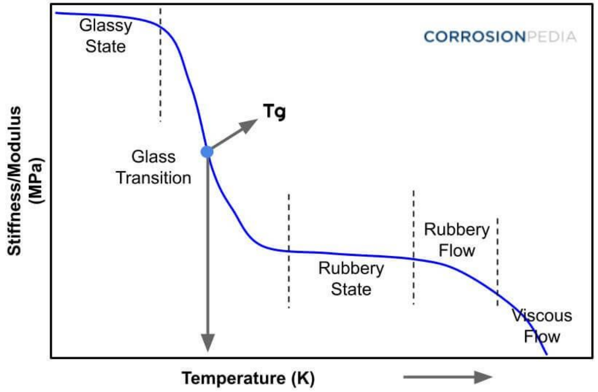


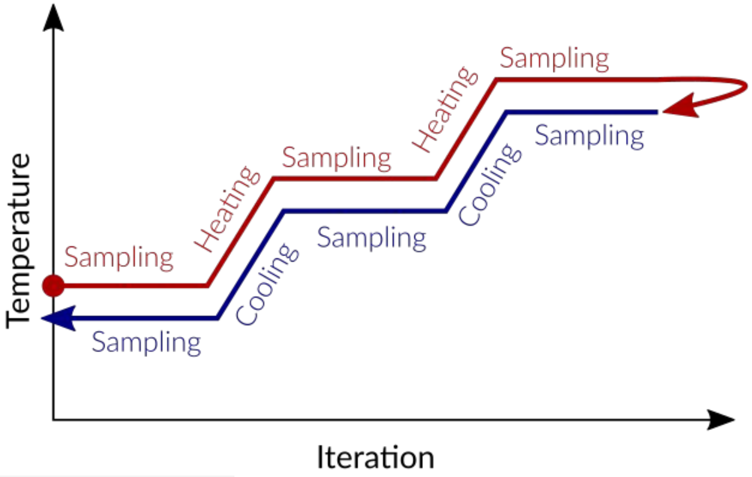




- **Glass transition temperature**
  - The glass transition temperature (Tg) is the crucial temperature range where an amorphous material, like a plastic or polymer, shifts from a hard, brittle, glassy state to a softer, rubbery, or leathery one as it's heated, due to increased molecular chain mobility.
  - Can be calculated using simulated annealing in GROMACS

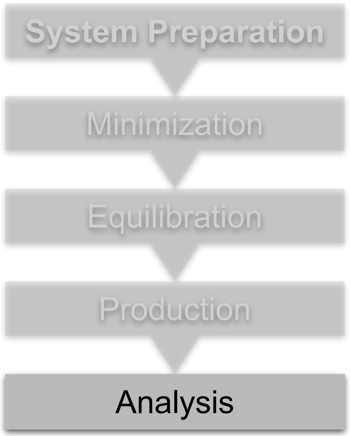


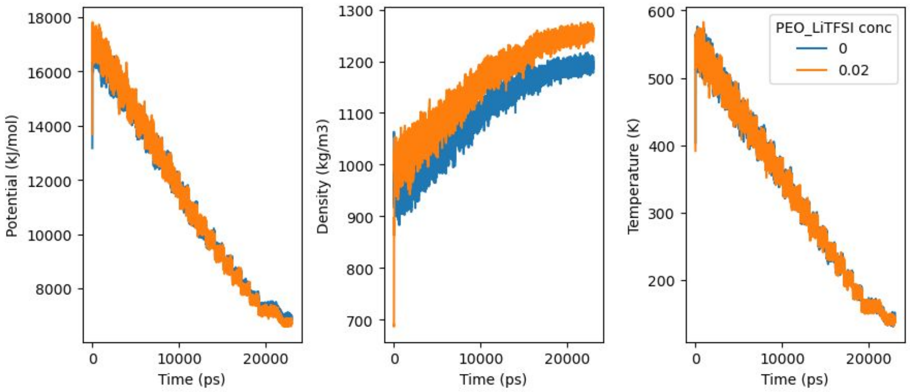


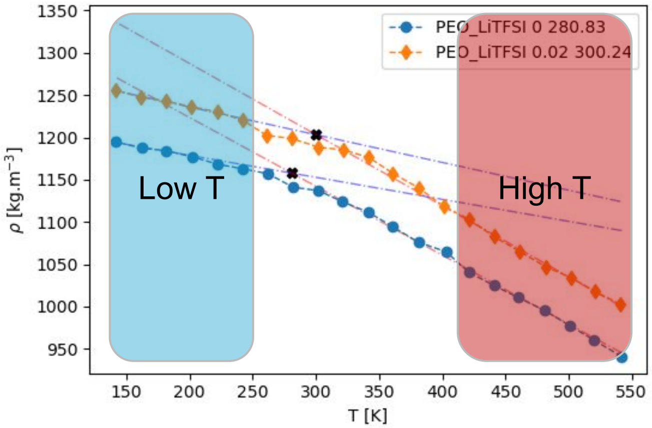



```
; Temperature coupling
tcoupl                  = V-rescale
tc-grps                 = System
tau_t                   = 0.1
ref_t                   = 400
annealing               = single
annealing_npoints       = 9
annealing_time          = 0 250 500 750 1000 1250 1500 1750 2000
annealing_temp          = 400 380 340 320 300 280 260 250 240
```

- **Glass transition temperature**
    - Example system - PEO polymer with LiTFSI salt (solid polymer electrolyte)

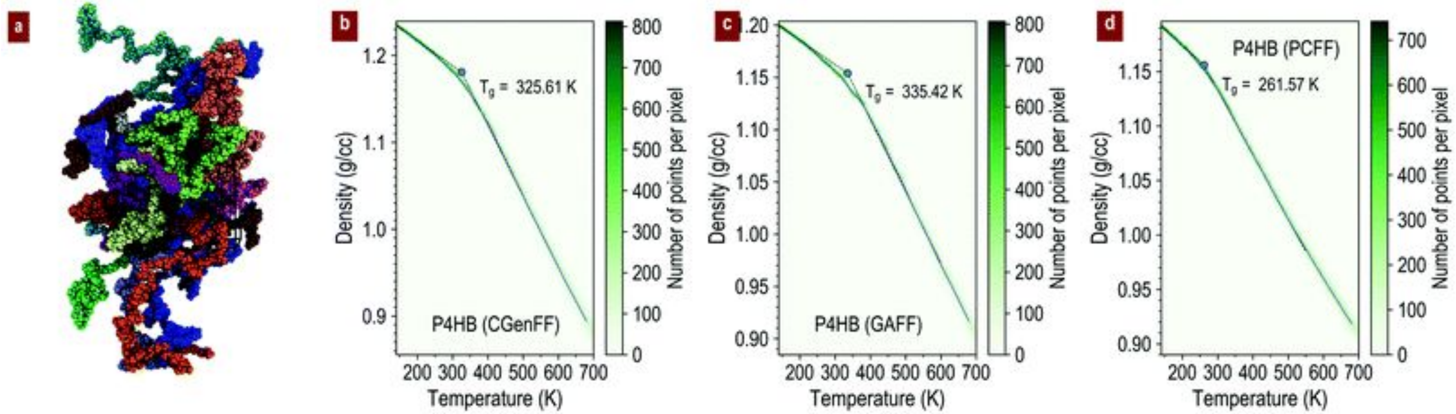


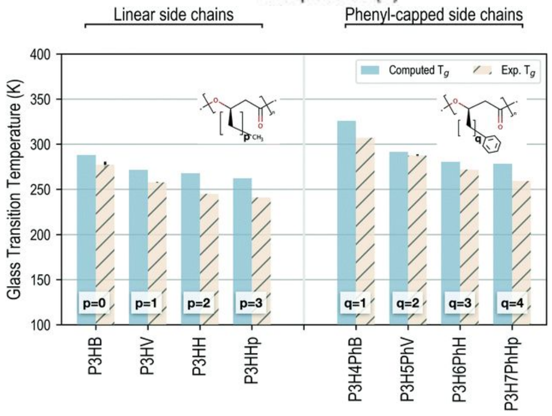




Tg is calculated as:
Intersection point of linear fits at higher temperature and low temperature regions

Tg of biopolymers from MD

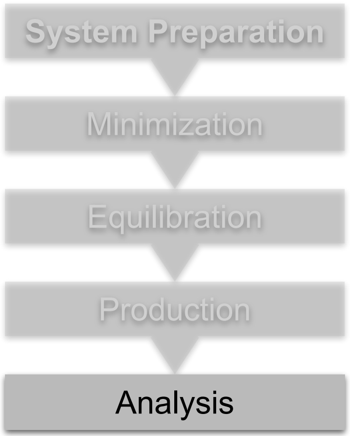




*"directly comparing the **computed Tg predictions** of various polymers with different chemistries, polymer side chain lengths and functional groups forming the polymer side chains against the respective **experimentally measured values**"*

Phys. Chem. Chem. Phys.,2020, 22, 17880

Other common properties



- H-bonding interactions: Either between polymer-water or polymer-polymer chains
    - *gmx hbond* allows using geometric definition of hydrogen bonds to define them throughout the structure.
- Mean Square Displacements (msd) and diffusion coefficients
    - *gmx msd*
- Mechanical properties from stress-strain curves. (Running Non-equilibrium MD simulations)